



GIRRAWEEEN HIGH SCHOOL

2016

MATHEMATICS EXTENSION 2

YEAR 12 Trial

HIGHER SCHOOL CERTIFICATE

EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section 1**10 marks****Attempt questions 1-10****Allow about 15 minutes for this section****Use the multiple-choice answer sheet for Questions 1-10.**

1. If $z = 1 + 2i$ and $w = 3 - i$, what is the value of $z - \bar{w}$?

- A) $3i - 2$ B) $4 + 3i$ C) $i - 2$ D) $4 + i$

2. What value of z satisfies $z^2 = 7 - 24i$?

- A) $4 - 3i$ B) $-4 - 3i$ C) $3 - 4i$ D) $-3 - 4i$

3. The equation of the polynomial equation $x^3 - 3x^2 + 2 = 0$ has roots α, β and γ . What is the value of $\alpha^3 + \beta^3 + \gamma^3$?

- A) 9 B) 13 C) 21 D) 25

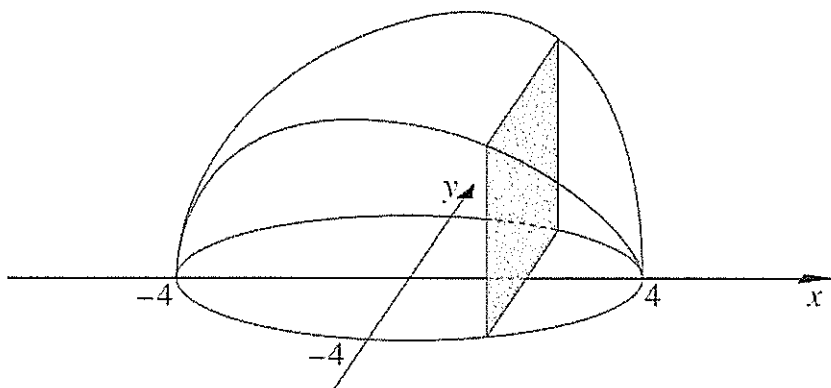
4. Which of the following is a correct expression for $\int x 3^{x^2} dx$?

- A) $\frac{3^{x^2+1}}{x^2+1} + C$ B) $\frac{3^{x^2}}{\ln 9} + C$ C) $\frac{3^{x^2}}{\ln 3} + C$ D) $3^{x^2} \ln 3 + C$

5. What is the eccentricity of the ellipse $9x^2 + 16y^2 = 25$?

- A) $\frac{7}{16}$ B) $\frac{\sqrt{7}}{4}$ C) $\frac{\sqrt{15}}{4}$ D) $\frac{5}{4}$

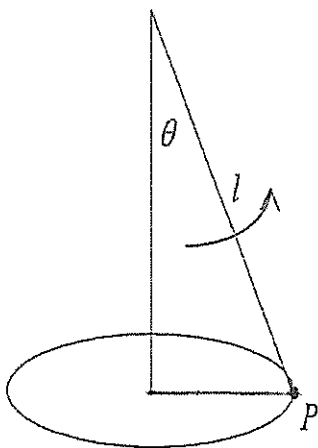
9. The base of a solid is the region bounded by the circle $x^2 + y^2 = 16$. Vertical cross-sections are squares perpendicular to the x -axis as shown in the diagram.



Which integral represents the volume of the solid ?

- A) $\int_{-4}^4 4x^2 dx$ B) $\int_{-4}^4 4\pi x^2 dx$ C) $\int_{-4}^4 4(16 - x^2) dx$ D) $\int_{-4}^4 4\pi(16 - x^2) dx$

10. A bob P of mass $m \text{ kg}$ is suspended from a fixed point A by a string of length l metres, and acceleration due to gravity g . P describes a horizontal circle with uniform angular velocity $\omega \text{ rad/sec}$.



Which of the following expressions represents the tension in the string ?

- A) $ml\omega$ B) $ml\omega^2$ C) $mg l \omega$ D) $mg l \omega^2$

Question 12. (15 marks).

a) The polynomial $P(x) = x^4 + ax^2 + bx + 28$ has a double root at $x = 2$.

What are the values of a and b ?

2

b) The polynomial $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$ has roots $a + ib$ and $a + 2ib$

where a and b are real and $b \neq 0$.

i) By evaluating a and b , find all the roots of $P(x)$.

3

ii) Hence, or otherwise, find the quadratic polynomials with real coefficients that are

factors of $P(x)$.

1

c) Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx$ for all integers $n \geq 0$.

i) Show that $I_n = \frac{1}{n-1} - I_{n-2}$ for integers $n \geq 2$.

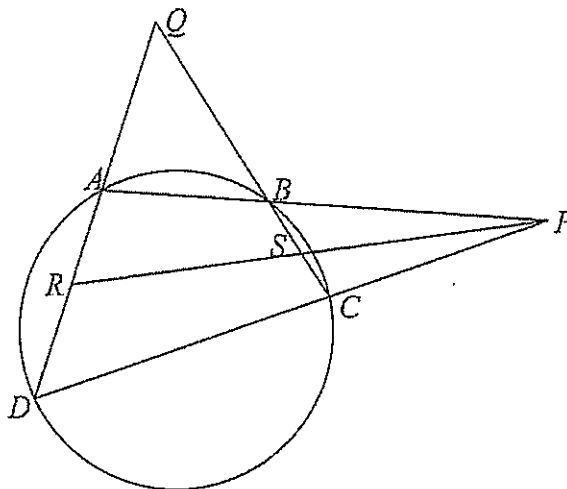
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ii) Hence find $\int_0^{\frac{\pi}{4}} \tan^5 x dx$.

2

Question 13. (15 marks)

- a) The diagram shows a cyclic quadrilateral $ABCD$. Chords AB and DC produced meet at P and chords DA and CB produced meet at Q . PR is the internal bisector of $\angle APD$ meeting AD at R and BC at S .



Prove that $\triangle QRS$ is isosceles.

2

- b) i) Show that $4x^2 + 9y^2 + 16x + 18y - 11 = 0$ represents an ellipse.

2

- ii) Find the eccentricity and hence the coordinates of its foci and the equations of the directrices.

2

Question 14.(15 marks)

a) Find the equation of the tangent to the curve $x^2 - xy + y^3 = 1$ at the point $P(1,1)$. 3

b) Use the substitution $x = \tan \theta$ to evaluate $\int_1^{\sqrt{3}} \frac{1}{x^2 \sqrt{1+x^2}} dx$. 3

c) i) Use De Moivre's Theorem to prove that if $z = \cos \theta + i \sin \theta$,

$$2 \cos n\theta = z^n + \frac{1}{z^n}. \quad \text{2}$$

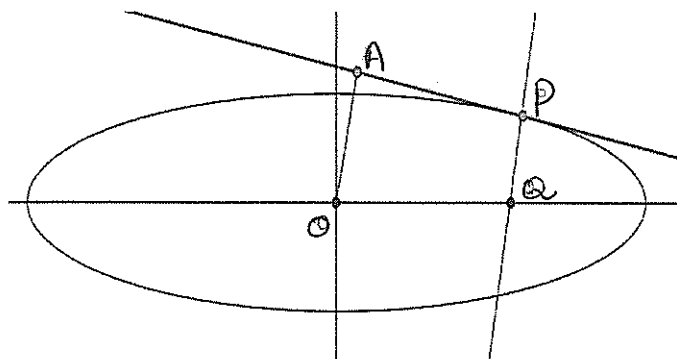
ii) Hence, or otherwise solve the equation $5x^4 - 11x^3 + 16x^2 - 11x + 5 = 0$ 3

d) A mass of 5 kilogram attached to a light fishing line describes a circular path with radius 60 centimetres about a point P on a smooth table. It completes 2 revolutions per second.

i) Find the tension in the fishing line. 2

ii) The line breaks under a tension of 900 Newtons. Find the maximum number of revolutions per second. 2

c) An ellipse has the equation $x^2 + 16y^2 = 25$.



i) Find the gradient of the ellipse at the point $P(3,1)$.

1

ii) Find the equation of the tangent and normal to the ellipse at P .

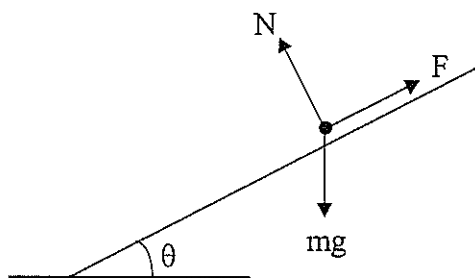
1

iii) The normal to the ellipse, at point P , meets the major axis at Q . A line from the centre, O to the tangent at P meets at right angles at point A .

Show that the value of $PQ \times OA$ is equal to the square of the semi-minor axis.

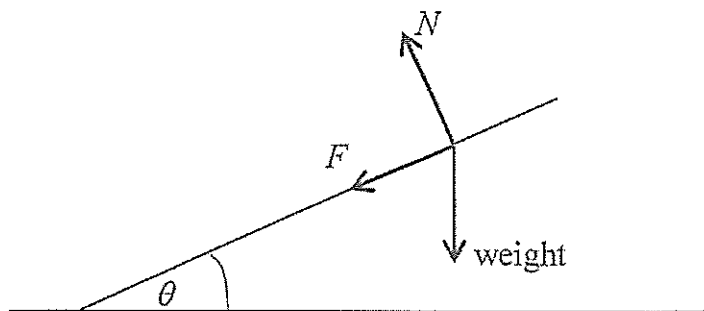
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c)



A particle of mass m is lying on an inclined plane and does not move. The plane is at an angle θ to the horizontal. The particle is subject to a gravitational force mg , a normal reaction force N , and a frictional force F parallel to the plane, as shown in the diagram. Resolve the forces acting on the particle, and hence find an expression for $\frac{F}{N}$ in terms of θ . 2

d)



A car of mass 2000 kg travels around a curve of radius 150m at a speed of 110 km/h.

The car experiences a lateral resistance force F of $0.22 N$, where the normal force is N , as shown in the diagram. By resolving the forces vertically and horizontally, find the angle θ for the car to negotiate the curve. (Assume acceleration due to gravity is $10m/s^2$). 4

END OF EXAMINATION

Year 12 Ext. 2 Solutions Trial 2016
Multiple Choice Questions

① $z = 1+2i$, $w = 3-i$, $z - \bar{w} = ?$

$$z - \bar{w} = 1+2i - (3+i) = 1+2i-3-i = -2+i \quad \text{(C)}$$

② $z^2 = 7-24i$

Let $z = a+ib$

$$\therefore z^2 = a^2 - b^2 + 2iab$$

$$\therefore a^2 - b^2 = 7$$

$$2ab = -24$$

By inspection: $a = \pm 4$, $b = \mp 3$

$$\therefore z = 4-3i$$

(A)

③ $x^3 - 3x^2 + 2 = 0$ has roots α, β, γ

$$\alpha^3 + \beta^3 + \gamma^3 = 3(\alpha^2 + \beta^2 + \gamma^2) - 6$$

$$= 3[(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)] - 6$$

$$= 3 \times 9 - 6 = 21$$

(C)

④ $\int x 3^{x^2} dx = \frac{1}{2} \int 2x 3^{x^2} dx = \frac{3^{x^2}}{2 \ln 3} + C = \frac{3^{x^2}}{\ln 9} + C$

(B)

Let $u = x^2$. $\therefore \int x 3^{x^2} dx = \frac{1}{2} \int 2x 3^{x^2} dx$

$$= \frac{1}{2} \int 3^u du = \frac{3^u}{2 \ln 3} = \frac{3^{x^2}}{\ln 9} + C$$

⑤

$$9x^2 + 16y^2 = 25 \Rightarrow$$

$$\frac{x^2}{\frac{25}{9}} + \frac{y^2}{\frac{25}{16}} = 1$$

$$b^2 = a^2(1-e^2) \Rightarrow \frac{25}{16} = \frac{25}{9}(1-e^2) \Rightarrow 9 = 16 - 16e^2$$

$$16e^2 = 7$$

$$e = \frac{\sqrt{7}}{4}$$

(B)

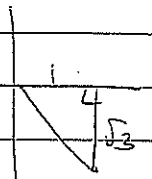
Multiple choice Questions

- ① C ② A ③ C ④ B ⑤ B
 ⑥ D ⑦ D ⑧ A ⑨ C ⑩ B

Question 11

a) (i) $z = 1 - i\sqrt{3}$
 $|z| = \sqrt{1^2 + (\sqrt{3})^2} = 2$

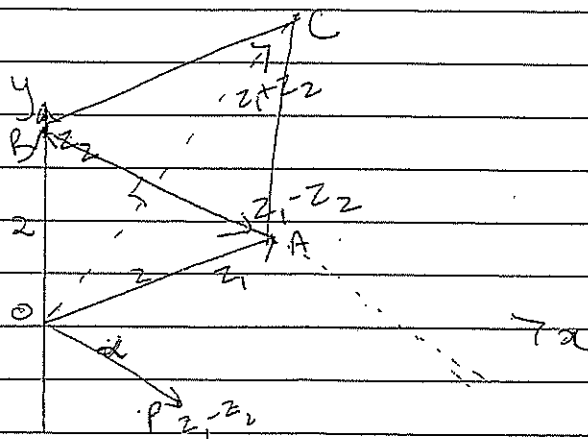
$\arg(z) = -\frac{\pi}{3}$



(ii) $z = 2 \left[\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right) \right]$

$z^6 = 2^6 \left[\cos(-2\pi) + i\sin(-2\pi) \right] = 2^6 [1 - 0] = 64$

b) (i)



(ii) Vectors OC and AB form a parallelogram.
 However, $OA = OB = 2$ $\therefore$ $OACB$ is a rhombus

$\angle AOC = \frac{1}{2} \left(\frac{\pi}{2} - \frac{\pi}{12} \right) = \frac{5\pi}{24}$

$\therefore \arg(z_1 + z_2) = \frac{5\pi}{24} + \frac{\pi}{12} = \frac{7\pi}{24}$

$$e) (i) \cot \theta + \operatorname{cosec} \theta = \cot(\theta/2)$$

$$\begin{aligned} \text{LHS } \cot \theta + \operatorname{cosec} \theta &= \frac{2 \cos^2 \theta}{2} \\ &= \frac{\cos \theta}{\sin \theta} + \frac{1}{\sin \theta} = \frac{\cos \theta + 1}{\sin \theta} \\ &= \frac{2 \cos^2 \theta/2}{2 \sin \theta/2 \cos \theta/2} = \cot \theta/2 = \text{RHS.} \end{aligned} \quad (2)$$

$$\begin{aligned} (ii) \int (\cot \theta + \operatorname{cosec} \theta) d\theta &= \int \cot \theta/2 d\theta \\ &= \int \frac{\cos \theta/2}{\sin \theta/2} d\theta = 2 \ln(\sin \theta/2) + C \end{aligned} \quad (+)$$

Question 12

$$a) P(x) = x^4 + ax^2 + bx + 28$$

$$P'(x) = 4x^3 + 2ax + b$$

$$\text{Root at } x=2 \Rightarrow$$

$$P(2) = 2^4 + 4a + 2b + 28 = 0$$

$$4 \cdot 4 + 4a + 2b = 0$$

$$4a + 2b = -44 \quad (1)$$

$$P'(2) = 0 \Rightarrow 32 + 4a + b = 0$$

$$4a + b = -32 \quad (2)$$

$$(1) - (2) \quad b = -12$$

$$\text{Sub'in (1)} \Rightarrow 4a - 24 = -44$$

$$\therefore 4a = -20$$

$$a = -5$$

$$a = -5$$

$$b = -12$$

(2)

Question 12 (c) continued

$$(ii) \quad \int_0^{\frac{\pi}{4}} \tan^5 x \, dx = I_5 = \frac{1}{4} - I_3$$

$$I_3 = \frac{1}{2} - I_1$$

$$\begin{aligned} I_1 &= \int_0^{\frac{\pi}{4}} \tan x \, dx = -[\ln(\cos x)]_0^{\frac{\pi}{4}} \\ &= -\ln \frac{1}{\sqrt{2}} = \frac{1}{2} \ln 2. \end{aligned}$$

$$\therefore I_3 = \frac{1}{2} - \frac{1}{2} \ln 2$$

$$\therefore I_5 = \frac{1}{4} - \left(\frac{1}{2} - \frac{1}{2} \ln 2 \right) = \frac{1}{2} \ln 2 - \frac{1}{4} \quad (2)$$

d) Solving simultaneously, $3 - x^2 = x + x^2$

(i)

$$2x^2 + x - 3 = 0 \Rightarrow (2x+3)(x-1) = 0$$

$$\therefore x = -\frac{3}{2}, x = 1$$

Since P is in first quadrant, $x = 1$ (1)

(ii)

$$\begin{aligned} \Delta x &= 2\pi r h \, \delta x \\ &= 2\pi (x+1) (y_1 - y_2) \, \delta x \end{aligned}$$

$$\begin{aligned} &= 2\pi (x+1) [3 - x^2 - x - x^2] \, \delta x \\ &= 2\pi (x+1) [3 - x - 2x^2] \, \delta x \end{aligned}$$

$$\therefore V = \lim_{\delta x \rightarrow 0} 2\pi \sum_{x=-1}^1 (x+1) (3 - x - 2x^2) \, \delta x \quad (2)$$

$$(iii) \quad V = 2\pi \int_{-1}^1 (3 + 2x - 3x^2 - 2x^3) \, dx = 8\pi \quad (1)$$

Question 13 continued

b) (i) $b^2 = a^2(1 - e^2)$

$$4 = 9(1 - e^2) \Rightarrow e^2 = \frac{5}{9}$$

$$\therefore e = \frac{\sqrt{5}}{3} \text{ as } (0 < e < 1)$$

directrices: $x = -2 \pm \frac{a}{e}$

$$\frac{a}{e} = \frac{3}{\frac{\sqrt{5}}{3}} = \frac{9}{\sqrt{5}}$$

focus:

$$ae = 3 \times \frac{\sqrt{5}}{3} = \sqrt{5} \Rightarrow (-2 + \sqrt{5}, -1) \text{ and } (-2 - \sqrt{5}, -1)$$

(2)

c) (i) $xy = c^2$

differentiating $\Rightarrow x \cdot \frac{dy}{dx} + y = 0$

$$\therefore \frac{dy}{dx} = -\frac{y}{x}$$

at P, $\frac{dy}{dx} = -\frac{c}{p} = -\frac{1}{p^2}$

$\therefore$ equation of tangent: $y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$

$$x - cp = -x + cp \Rightarrow x + p^2 y = 2cp \quad (1)$$

(ii) Tangents at P and Q are

$$x + p^2 y = 2cp \quad (1)$$

$$x + q^2 y = 2cq \quad (2)$$

$$(1) - (2) \quad y(p^2 - q^2) = 2c(p - q)$$

$$\therefore y = \frac{2c}{p+q}$$

Sub. into (1) $\Rightarrow x + p^2 \left(\frac{2c}{p+q} \right) = 2cp$

$$\therefore T = \left(\frac{2cpq}{p+q}, \frac{2c}{p+q} \right) \quad x = 2cp - \frac{2cp^2}{p+q} = \frac{2cpq}{p+q}$$

Question 14

a) $x^2 - xy + y^3 = 1$ $P(1,1)$

differentiating, $2x - x \cdot \frac{dy}{dx} - y + 3y^2 \frac{dy}{dx} = 0$

$$\frac{dy}{dx} (3y^2 - x) = y - 2x$$

$$\therefore \frac{dy}{dx} = \frac{y - 2x}{3y^2 - x}$$

at $P(1,1)$, $\frac{dy}{dx} = \frac{1 - 2}{3 - 1} = \frac{-1}{2} = -\frac{1}{2}$

eqⁿ of tangent:

$$y - 1 = -\frac{1}{2}(x - 1)$$

$$2y - 2 = -x + 1 \Rightarrow x + 2y - 3 = 0$$

(3)

b) $x = \tan \theta \quad \therefore dx = \sec^2 \theta d\theta$

Limits: $x = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}$

$$x = 1 \Rightarrow \theta = \frac{\pi}{4}$$

$$x^2 \sqrt{1+x^2} = \tan^2 \theta \cdot \sec \theta = \tan^2 \theta \cdot \frac{\sin \theta}{\cos \theta}$$

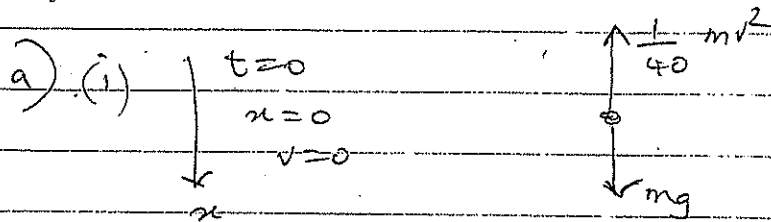
$$\therefore = \tan^2 \theta \cdot \sin \theta \sec \theta$$

$$\therefore \int_1^{\sqrt{3}} \frac{1}{x^2 \sqrt{1+x^2}} dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \operatorname{cosec} \theta \cot \theta d\theta$$
$$= -[\operatorname{cosec} \theta]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$

$$= -\frac{2}{\sqrt{3}} + \sqrt{2} = \frac{1}{3}(3\sqrt{2} - 2\sqrt{3})$$

(3)

Question 15



By Newton's second law, $m\ddot{x} = mg - \frac{1}{40}mv^2$

$$\therefore \ddot{x} = \frac{1}{40}(400 - v^2)$$

(1)

(ii)

$$\ddot{x} = \frac{1}{40}(400 - v^2)$$
$$\frac{dv}{dt} = \frac{1}{40}(400 - v^2)$$

$$\therefore \frac{dt}{dv} = \frac{40}{20^2 - v^2} = \frac{1}{20+v} + \frac{1}{20-v}$$

$$\therefore \int dt = \int \frac{1}{20+v} dv + \int \frac{1}{20-v} dv$$

$$\therefore t = \ln\left(\frac{20+v}{20-v}\right) + C$$

when $t=0$, $v=0 \Rightarrow C=0$.

$$\therefore t = \ln\left(\frac{20+v}{20-v}\right)$$

(2)

(iii)

$$\therefore e^t = \frac{20+v}{20-v}$$

$$\therefore (20-v)e^t = 20+v = 20e^t - ve^t$$

$$\therefore v(1+e^t) = 20(e^t-1)$$

$$\therefore v = \frac{20(e^t-1)}{1+e^t} = 20\left(\frac{e^t+1-2}{e^t+1}\right)$$

$$= 20\left(1 - \frac{2}{e^t+1}\right)$$

(1)

b) (i) $h = AF = a(e-1)$

$\therefore PF = a(e^2-1) = h(e+1)$

c) (i) $x^2 + 16y^2 = 25$

differentiating, $2x + 32y \cdot \frac{dy}{dx} = 0$

$\therefore \frac{dy}{dx} = \frac{-2x}{32y} = \frac{-x}{16y}$

at $P(3,1)$, $\frac{dy}{dx} = -\frac{3}{16}$ (1)

(ii) Equation of tangent:

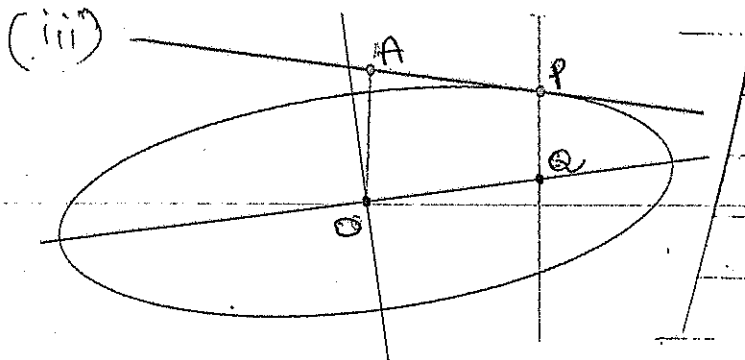
$y-1 = -\frac{3}{16}(x-3) \Rightarrow 16y-16 = -3x+9$

$\therefore 3x+16y=25$

Equation of normal:

$y-1 = \frac{16}{3}(x-3) \Rightarrow 3y-3 = 16x-48$

$\therefore 16x-3y=45$



$OA \perp AP$ and
eqⁿ AP $\Rightarrow 3x+16y-25=0$

$\therefore OA = \left| \frac{0+0-25}{\sqrt{9+256}} \right| = \frac{25}{\sqrt{265}}$

Coordinates of Q $\Rightarrow$ sub. in $y=0$ in $16x-3y=45$

$\therefore Q = \left(\frac{45}{16}, 0 \right)$ $\therefore x = \frac{45}{16}$

$\therefore PQ = \sqrt{\left(3 - \frac{45}{16}\right)^2 + 1} = \sqrt{\frac{9}{256} + 1} = \frac{\sqrt{265}}{16}$

$\therefore PQ \times OA = \frac{25}{\sqrt{265}} \times \frac{\sqrt{265}}{16} = \frac{25}{16}$

but in $x^2 + 16y^2 = 25 \Rightarrow \frac{x^2}{25} + \frac{y^2}{\frac{25}{16}} = 1 \Rightarrow b^2 = \frac{25}{16}$

$\therefore PQ \times OA = \frac{25}{16} = \text{square of semi-minor axis}$

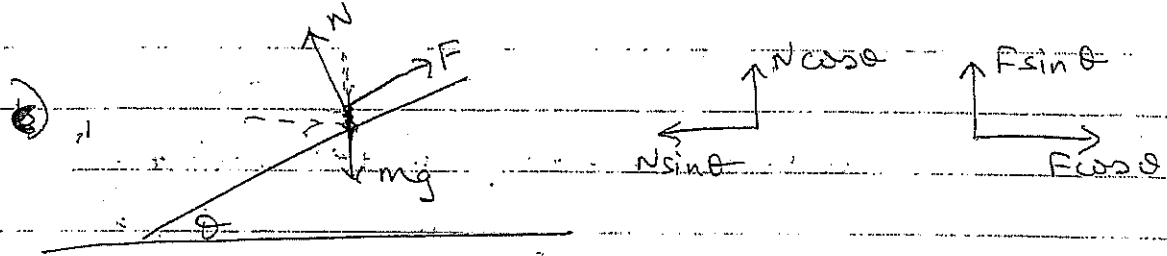
$$(iv) \quad \textcircled{1} \quad x \sin \beta + 2x \cos \beta$$

$\Rightarrow$

$$T_1 (\sin \beta \cos \alpha + \cos \beta \sin \alpha) = mg \sin \beta + mr \omega^2 \cos \beta$$

$$\therefore T_1 = \frac{M (r \omega^2 \cos \beta + g \sin \beta)}{\sin(\alpha + \beta)}$$

$\textcircled{2}$

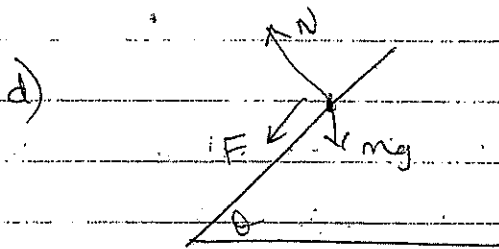


resolving forces $\therefore$

normal to plane: $N = mg \cos \theta$

parallel to plane: $F = mg \sin \theta$

$$\therefore \frac{F}{N} = \frac{mg \sin \theta}{mg \cos \theta} = \tan \theta \quad \textcircled{2}$$



Acceleration

$$\frac{v^2}{r}$$

$$110 \text{ km/h} = \frac{275}{9} \text{ m/s}$$

Resolve vertically $\Rightarrow mg + F \sin \theta = N \cos \theta$

Resolve horizontally $\Rightarrow F \cos \theta + N \sin \theta = \frac{mv^2}{r}$

$F = 0.22N, \Rightarrow mg + 0.22N \sin \theta = N \cos \theta$

$$N(\cos \theta - 0.22 \sin \theta) = 20000$$

$$N(0.22 \cos \theta + \sin \theta) = 12448.56$$

Dividing,

$$\frac{\cos \theta - 0.22 \sin \theta}{0.22 \cos \theta + \sin \theta} = 1.6066 \dots$$

$$\therefore \frac{1 - 0.22 \tan \theta}{0.22 + \tan \theta} = 1.6066 \Rightarrow \tan \theta = 0.3539$$

$$\therefore \theta = 19.29^\circ$$